

What is physics?

THIS FIELD OF SCIENCE SEEKS TO REVEAL THE WORKINGS OF THE UNIVERSE ON THE LARGEST AND SMALLEST SCALES

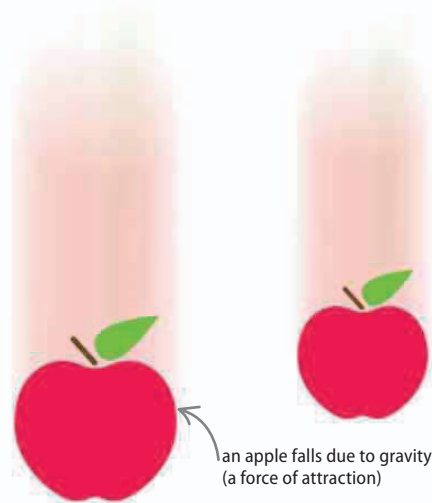
The word “physics” means “nature” in ancient Greek, and physicists tackle the most fundamental subjects in the Universe, such as the nature of energy, space, and even time.

Foundation of knowledge

Physics is the foundation of all scientific knowledge. Chemistry, biology, and other sciences are built on an understanding of physics. For example, physicists have revealed the structure of the atom, which chemists use to understand how chemicals react with each other. Meanwhile, physics has also explained how energy behaves, which is crucial knowledge for biologists figuring out how organisms stay alive. A few physicists, such as Albert Einstein and Isaac Newton, have become famous because their discoveries have had such a far-reaching effect.

▷ Falling objects

Physics explains many everyday phenomena. For example, Newton's theory of gravitation (see pages 178–179) explains why an apple—or any object—falls to the ground.



Energy, mass, space, and time

Physics can express everything in the Universe in terms of mass, energy, and force, from the workings of a giant star to a raindrop falling from a cloud. A mass is an object that is affected by forces. What a force does is transfer energy from one mass to another, which changes the way the masses move or are shaped. For example, throwing a ball or stretching a rubber band requires force—even light shining on an object exerts a tiny force on it!

◁ In motion

They may not know it, but basketball players use physics. They push the ball in just the right direction and with just the right force to score a basket.

